

```
In [1]: import numpy as np
array=np.random.randint(1,100,9)
array
```

```
Out[1]: array([39, 86, 40,  1, 99, 41,  8, 42, 97], dtype=int32)
```

```
In [2]: np.sqrt(array)
```

```
Out[2]: array([6.244998 , 9.2736185 , 6.32455532, 1.          , 9.94987437,
        6.40312424, 2.82842712, 6.4807407 , 9.8488578 ])
```

```
In [3]: array.ndim
```

```
Out[3]: 1
```

```
In [4]: new_array=array.reshape(3,3)
new_array
```

```
Out[4]: array([[39, 86, 40],
        [ 1, 99, 41],
        [ 8, 42, 97]], dtype=int32)
```

```
In [5]: new_array.ndim
```

```
Out[5]: 2
```

```
In [6]: new_array.ravel()
```

```
Out[6]: array([39, 86, 40,  1, 99, 41,  8, 42, 97], dtype=int32)
```

```
In [7]: newm=new_array.reshape(3,3)
```

```
In [8]: newm
```

```
Out[8]: array([[39, 86, 40],
        [ 1, 99, 41],
        [ 8, 42, 97]], dtype=int32)
```

```
In [9]: newm[2,1:3]
```

```
Out[9]: array([42, 97], dtype=int32)
```

```
In [10]: newm[1:2,1:3]
```

```
Out[10]: array([[99, 41]], dtype=int32)
```

```
In [11]: new_array[0:3,0:0]
```

```
Out[11]: array([], shape=(3, 0), dtype=int32)
```

```
In [12]: new_array[0:2,0:1]
```

```
Out[12]: array([[39],  
               [ 1]], dtype=int32)
```

```
In [13]: new_array[0:3,0:1]
```

```
Out[13]: array([[39],  
               [ 1],  
               [ 8]], dtype=int32)
```

```
In [14]: new_array[1:3]
```

```
Out[14]: array([[ 1, 99, 41],  
               [ 8, 42, 97]], dtype=int32)
```

```
In [ ]:
```